Writing a good protocol

This is a bench card. Full protocol available online.

›› *Writing a protocol*

(1.) Sketch the structure first. At minimum: a unique identifier, a name, an abstract, a changelog, and one or more procedures. Add safety, analysis, or troubleshooting advice where applicable. 🕮

(2.) Write each step as a single action in active voice.

(3.) Be deliberate:

- Name equipment, reagents, and durations a new reader would need to execute the step unassisted. Assume nothing about prior knowledge — including your own, two years from now.
- Parameterize quantities, volumes, and ratios so the procedure scales. A protocol is not a lab notebook. Leave room for the next user to adjust without rewriting it.
- When an action is non-obvious or grounded in experience, attach a one-sentence rationale. 🕮

(4.) *Critical:* Mark critical steps and decision points so they are visually distinct from routine ones. Readers scan; the moments that matter most should be impossible to skim past.

(5.) Track changes:

- Maintain a changelog. Append one line per substantive edit with the date, the editor's full name, and a one-sentence summary of what changed. 🕮
- Version the filename. For incompatible changes, create a new file with an incremented version number and leave the retired version intact for reproducibility.
- *Optional:* Commit changes through version control 🗎 SOP 0101. 🕮

(6.) Give credit:

- Audit AI-assisted content. If a language model drafted or revised any section, verify every factual claim and citation before publishing. Remove anything that can't be confirmed.
- Do not copy-paste copyright-protected material without permission or a compatible license. When adapting content from another protocol, cite it and confirm its license is compatible with the one intended for publication.

(7.) Review protocols regularly with users and collaborators. A second reader catches assumptions the writer is blind to.

🕮 Notes (available online)

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